Volatility-Conditioned Mean Reversion

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Overview. This project evaluates whether mean-reversion trading signals perform better under specific market conditions. The initial hypothesis was that mean reversion would be more reliable in low-volatility environments, where price movements are more stable and less noisy. However, the analysis ultimately revealed the opposite: mean reversion is empirically stronger during high-volatility periods. Using SPY (S&P 500 ETF) from 2014–present, with validation on AAPL, MSFT, NVDA, and JPM, the goal is to understand how market conditions affect signal performance.

Data and Setup. Daily adjusted closing prices were obtained from Yahoo Finance and cleaned by standardizing formats, checking for duplicates, and removing invalid or missing observations. To capture changing market conditions, volatility was estimated using a GARCH(1,1) model, which reflects the tendency for volatility to cluster over time. Each day was then classified into low, medium, or high volatility regimes.

Signal and Evaluation. Mean-reversion signals were constructed using a 20-day z-score to measure how far price deviates from its recent average. Signals were defined as buying when the z-score falls below -1 and selling when it exceeds 1. Performance was evaluated using forward returns over 1, 3, and 5 days, comparing all signals against those restricted to volatility regimes.

Results. Contrary to the initial hypothesis, filtering by high volatility meaningfully improves performance. For SPY at the 5-day horizon, the baseline strategy produces approximately zero return, while the high-volatility filtered strategy yields about +0.10% per trade, with hit rate improving from roughly 46% to 51%. The cross-asset check broadly supports this result, especially at 1-day and 3-day horizons, while 5-day results are more mixed. Overall, stronger price deviations tend to produce more reliable outcomes, suggesting that large, rapid price movements create temporary dislocations that are more likely to revert.

Extension: Contrarian Portfolio Approach. After testing simple volatility filtering, the analysis was extended using a cross-sectional contrarian strategy inspired by prior literature [1]. Instead of generating independent signals for each asset, this approach constructs a dollar-neutral portfolio that systematically buys recent relative losers and shorts recent relative winners across assets. This better reflects how mean-reversion strategies are implemented in practice.

Consistent with earlier findings, applying a high-volatility filter to this portfolio improves performance, suggesting that the effect is not limited to a single-asset signal but persists in a more realistic portfolio setting. This progression from single-asset signals to cross-sectional portfolios highlights the robustness of the volatility-conditioned mean-reversion effect.

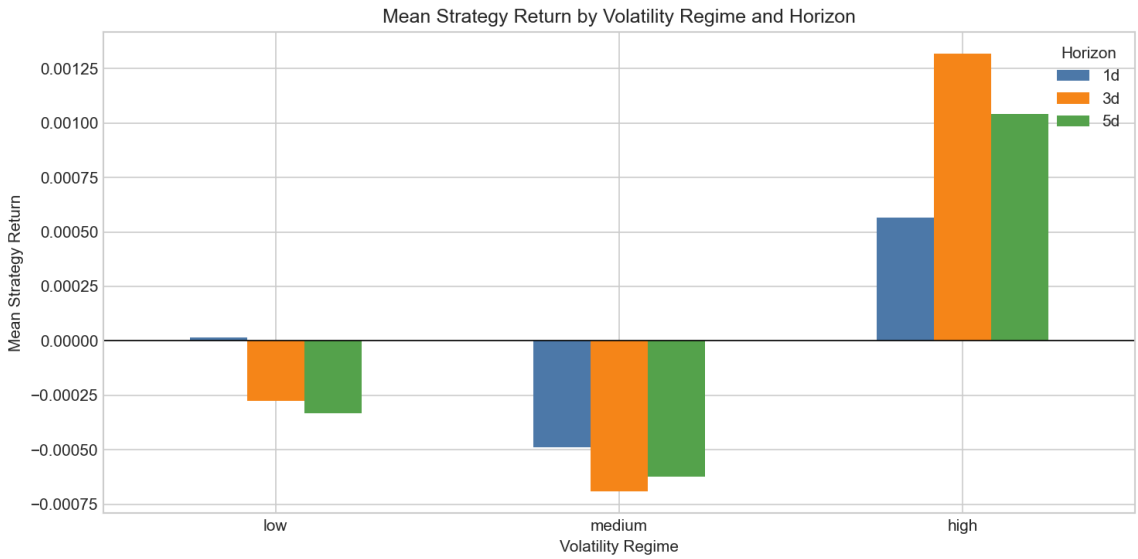

Figure 1: Mean-reversion performance is strongest in high-volatility regimes across multiple horizons.

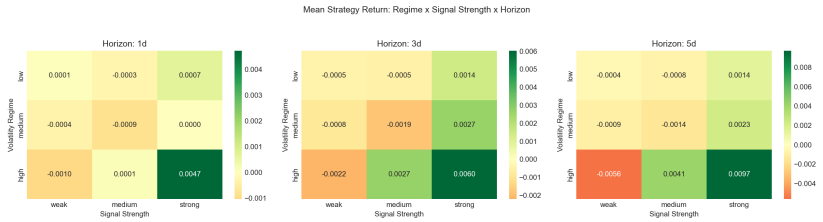

Figure 2: Performance improves further when combining high volatility with stronger signal deviations.

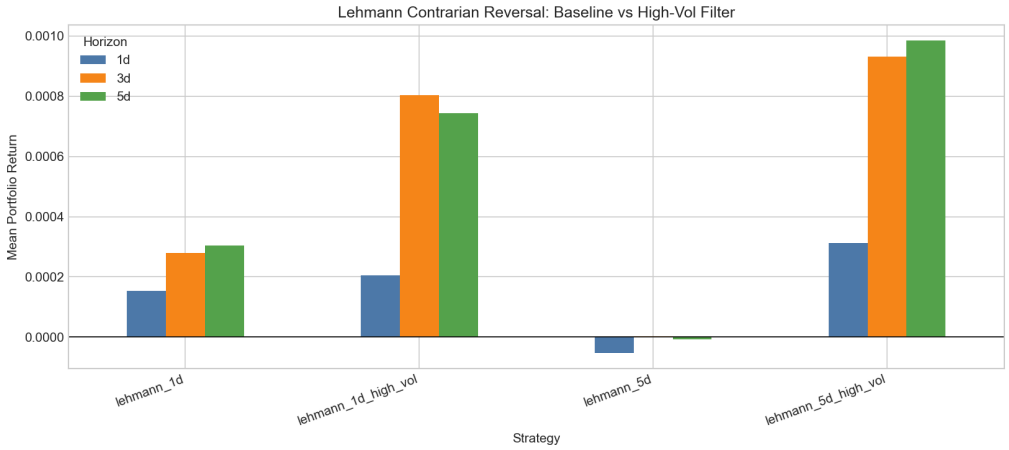

Figure 3: Lehmann-style contrarian portfolio: high-volatility filtering improves portfolio-level mean-reversion performance.

Conclusion and Reflection. Overall, the results show that market context plays a critical role in strategy performance. While it seemed reasonable that stable markets would produce more predictable reversions, the data suggests that turbulent markets create stronger opportunities due to larger and more frequent mispricings. These results should be interpreted as descriptive evidence rather than a live trading backtest, since regimes are estimated on the full sample and transaction costs are excluded. Future work includes walk-forward testing and incorporating transaction costs for a more realistic assessment.

References

[1] Bruce N. Lehmann. *Fads, Martingales, and Market Efficiency.* National Bureau of Economic Research, Working Paper No. 2533, 1990.